

Web2Music: Deployment-Level Measurement of Page-Conditioned Generative Audio in a Browser Extension

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Abstract—Generative audio models can produce plausible music from text, but placing one inside an always-on browser extension raises problems audio quality alone does not solve: deriving a prompt from a page, keeping browsing history private, and hiding a slow generator from the critical path. We present Web2Music and measure where these assumptions fail. A four-signal page descriptor adds well under a tenth of a second per page. A three-stage cascade (keyword, zero-shot NLI, LLM) barely moves accuracy across its confidence thresholds and, with a hosted middle stage, never meaningfully reduces off-device disclosure. Fallback-then-swap decouples audible start from generation, though some cold requests still run long enough to time out. Across a corpus of real sites, requests and generated clips we report five further negative results: a crossfade that worsens a large share of seams; substantial loss of the requested clip duration; a sensitive-content detector that misses most of an adversarial slice; weak mood agreement among annotators; and an engineered prompt that loses to a bare mood word on nearly every loop metric.

Index Terms—generative audio, context-aware computing, browser extensions, zero-shot classification, model cascades, on-device inference, ambient interfaces

I. INTRODUCTION

Background music is chosen once, for a session, and then ignored: a lo-fi playlist runs for four hours across a tax form, a horror wiki and a birthday message without acknowledging that anything changed. It is context-free because making it context-aware has historically required a human curator or a library covering every context in advance. Text-conditioned generative audio models remove that constraint in principle (§II): if the description can be derived from what the user is doing, the curator is no longer needed.

Deriving it is where the engineering lives, via three constraints, none about audio quality. *The prompt has to come from somewhere*: a page is a DOM of navigation chrome, ads and colour, not a prompt. *Classification is a privacy decision*: the most accurate way to classify a page is an LLM call, but doing that on every page turns a music player into a browsing-history exfiltration channel. *The model is too slow for the critical path*: on a CPU laptop the three cold requests that completed took a median of 139 s for a 10–20 s clip, and every 28 s cold request exceeded a 300 s client timeout, against a page load of about two seconds – a gap CPU inference does not close; the interaction must be designed so it is invisible.

This paper’s contributions follow the same three constraints. First, a four-signal page descriptor (text, sentence embedding, page colour, browsing behaviour) with its extraction cost measured over 105 real websites; an offline ablation under the keyword stage bounds colour’s effect on the mood label at ≤ 7.5 percent of pages and is uninformative for the other two auxiliary signals (§V-D).

Second, an abstaining three-stage classifier (free keyword heuristic $\rightarrow$ zero-shot NLI $\rightarrow$ remote LLM) and the finding that, with a hosted cheap stage, such a cascade *redistributes* disclosure between providers rather than reducing it: the thresholds are a privacy mechanism only if the middle stage is local, which we measure only on an 18-page subset.

Third, a latency-hiding playback design: instant fallback followed by an abort-on-stale crossfaded swap, with its full latency distribution and failure modes, including 12 of 144 client timeouts; and loop synthesis from a non-loop model, bar-snapped chroma self-similarity, a 50 ms crossfade, and a gapless Ogg/Opus container, evaluated over 141 clips.

TABLE I
CLOSEST RELATED SYSTEMS.

Authors	Approach	Pros	Cons
Copet et al. [1]	MusicGen: single-stage transformer LM over En-Codec tokens	• Open; small variant deployable	• One-shot clip, no loop closure • No context derivation
Agostinelli et al. [3]	MusicLM: hierarchical semantic/acoustic tokens	• Higher fidelity, long-form coherence	• Heavy; not openly deployable • Non-looping
Liu et al. [5]	AudioLDM: latent diffusion on CLAP embeddings	• Faster than autoregressive tokens	• Cost scales with sampling steps • No loop, no affect target
Agres et al. [6]	AffectMachine-Pop: rule-based generation from a valence–arousal target	• Truly real time, no GPU • Interpretable	• Affect target supplied externally, not derived from context
Liu et al. [17]	MetaBGM: two-stage LLM, game state → music description	• Fully automated • Multi-scene transitions	• Full state to a cloud LLM every call • No listener study
Suzawa et al. [18]	Discussion Jockey 2: Whisper + GPT-4 descriptions → MusicGen	• Live context • User study raised relaxation	• Full transcript to GPT-4 unfiltered • Small study
Wei et al. [19]	Context-AI Tunes: vision-LLM on a photographed scene + self-report	• Rigorous validation ($n = 26$)	• Scene image leaves the device • Active input
Chong et al. [20]	Casper: client-side rules → NER → LLM prompt sanitiser	• Fully client-side • Disclosure is the goal	• Filters only, never generates • No corpus exposure rate
Chen et al. [8]	FrugalGPT: confidence-gated cascade before expensive LLMs	• Cascade pattern this paper adopts	• Optimises cost/latency only • Never reports disclosure
Web2Music (this paper)	Passive page context → 3-stage abstaining classifier → MusicGen behind fallback-then-swap	• Passive, not a request • Disclosure measured as a rate • Latency and failure modes measured	• Perceptual benefit untested • Hosted cascade still discloses most pages

II. RELATED WORK

Generative audio. MusicGen [1] is a single-stage transformer language model over discrete EnCodec [2] tokens, text-conditioned and openly deployable, which is why its small variant is the generator here; MusicLM [3] and AudioLDM [5] trade fidelity against speed, and AudioGen [4] applies the same conditioning to environmental sound. All four emit a fixed-length one-shot excerpt whose end does not join its beginning. AffectMachine-Pop [6] is rule-based and truly real time, but takes its valence–arousal target [7] from outside rather than from context.

Context-aware music and disclosure. Four recent systems derive music from a user’s situation: MetaBGM [17] from game state, Discussion Jockey 2 [18] from live meeting speech, Context-AI Tunes [19] from a photographed scene plus self-reported stress, and, on the privacy side, Casper [20] sanitises a typed prompt with a client-side cascade before it reaches a web LLM. FrugalGPT [8] established the confidence-gated cascade pattern this paper’s classifier follows, framed as cost and latency optimisation. Table I places these against Web2Music on the properties this paper measures.

Web2Music combines three properties no recent system in Table I combines. *Passive context, not a request:* unlike MetaBGM, Discussion Jockey 2 and Context-AI Tunes, which respond to something a user is actively doing, Web2Music’s context is whatever page is open, sensed without prompting. *Disclosure as a measured rate:* every system in Table I sends its full context to a cloud model and reports no on-device fraction; Web2Music still discloses most pages as measured (§V-H), but the number is reported rather than assumed away. *Deployment failure modes as findings:* none of the four context-to-music papers reviewed reports a latency distribution, cache behaviour, loop-continuity measurement or adversarial audit. The contribution is accordingly the de-

ployment measurement – not a better generator, classifier or cascade, all three adopted from existing work – and the finding that several resulting guarantees do not hold as configured.

III. RESEARCH GAPS AND PROBLEM STATEMENT

A. Gaps

Prior work leaves four gaps, none about audio quality. *Deployment-level properties go unmeasured:* prompt cost, generation-time behaviour and safeguard robustness fall outside the audio-quality frame prior work uses. *On-device classification is treated as a cost problem, not a disclosure one:* no work we reviewed reports an *exposure rate* as a tunable curve. *Context-aware music systems assume a library:* whether generation beats retrieval on cost is asserted, not measured. *Generative audio models do not produce loops:* how much a crossfade removes is not characterised in the literature we found.

B. Problem statement

Given a rendered page p , produce ambient audio $a(t)$ appropriate to p , subject to *latency* (audible start bounded by the confidence window plus under a second, independent of generation time), *disclosure* (text leaving the device is measurable and zero for sensitive pages), *continuity* (a clip repeats without an audible seam), and *restraint* (identifiable page classes get no output, verifiably).

Five verdicts: fallback-then-swap keeps time-to-audio within budget (*supported*, §V-E); generation covers a request space retrieval does not (*argued only under a random-library model*, §V-L); loop synthesis reduces seam discontinuity (*gap-less achieved, seam reduction marginal*, §V-I); restraint policies fire correctly (*mechanism passes, detection fails*, §V-K); the cascade reduces exposure at a measurable cost (*negative for the hosted backend*, §V-H).

IV. SYSTEM ARCHITECTURE

Web2Music is a Manifest V3 Chrome extension plus one HTTP audio-generation service. Fig. 1 follows one page from DOM to speaker in nine numbered steps and marks the two persistent stores where they are touched: an IndexedDB vector store on the device and a Postgres cache index behind the generation service. The division of labour follows browser security boundaries: the descriptor extractor runs in the content script to reach the DOM, the classifier runs in the service worker since the content script’s origin is untrusted, and playback and ONNX inference run in an offscreen document since Manifest V3 forbids audio in a service worker and a page’s CSP blocks WebAssembly there. Three components are network services: the hosted zero-shot backend, the remote LLM, and the generation service.

A. Page descriptor extraction

A single function assembles the descriptor from four contextual signals – text, a 384-dimensional MiniLM sentence embedding (in-browser ONNX), colour (an area-weighted HSL histogram over up to 800 elements [9]), and behaviour (scroll/cursor velocity) – plus readability [10] as an auxiliary feature. The embedding is searched by cosine similarity against an IndexedDB vector store of prior pages (step 2 in Fig. 1) to flag a revisit, then upserted, capped at 2,000 entries; this search is the vector-store row of Table III. The embedding feeds the vector store only, not the keyword stage’s mood computation (§V-D depends on this). Extraction stays off the critical path: debounced, idle-scheduled, and re-run on navigation and DOM mutation.

B. The classifier cascade

The classifier maps the descriptor onto a 13-way category and 11-way mood, then a music profile and prompt. Category is a content-topic label; mood is the affective target that drives the profile, and is weakly agreed upon among annotators (§V-B).

Keyword stage. A keyword map gives the category, keyword-only at this stage; mood blends the keyword score with a colour bias and a behaviour bias. At confidence ≥ 0.5 the pipeline stops here at essentially zero cost; skipped by construction for non-English pages. **Zero-shot stage.** Scored against thirteen category hypotheses, accepting a label only if the top score is ≥ 0.45 and the margin over the runner-up is ≥ 0.10 , else abstaining. A hosted entailment backend transmits page text and is what every full-scale number here was measured with; a local quantised ONNX backend transmits nothing, measured only on 18 pages. **LLM stage.** An eight-second-timeout hosted model, falling back to the keyword stage on failure; the fraction of pages reaching it is the LLM-specific exposure rate, and total off-device rate adds the pages the zero-shot stage also received.

A committed mood maps to a music profile and a prompt. Every page the cascade-ablation harness processes is logged with every stage’s output, not just the winner, so §V-H can replay any configuration offline. A newly classified mood must

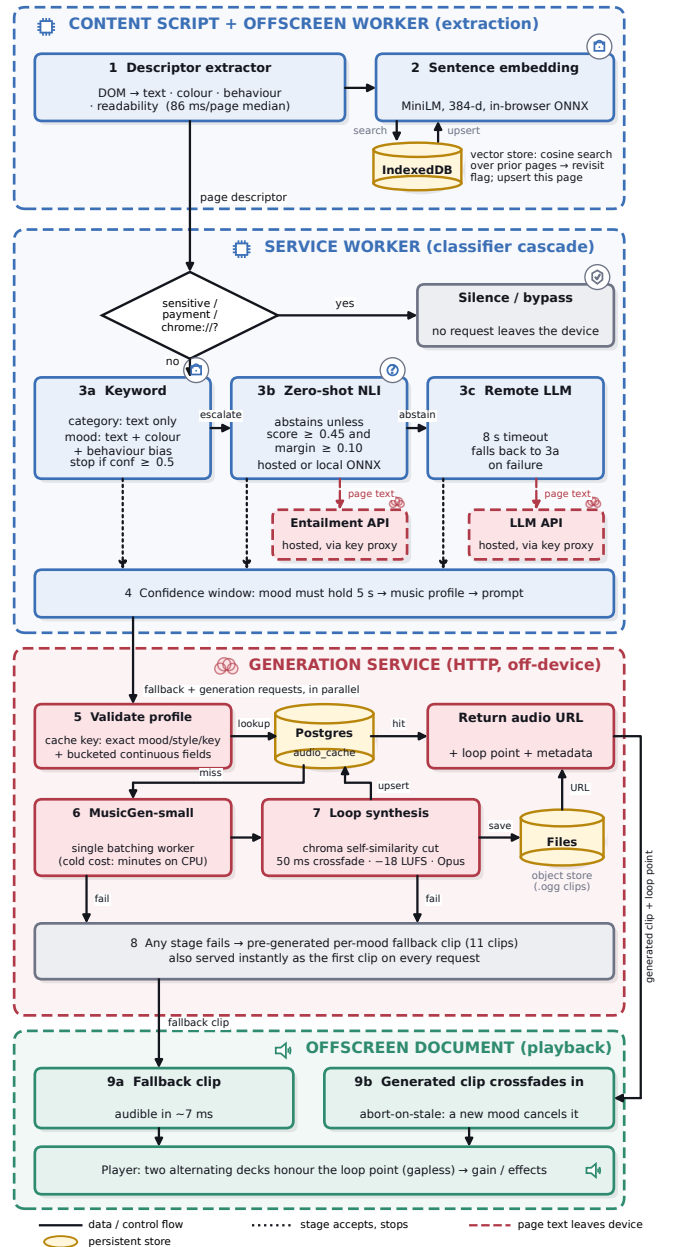


Fig. 1. System architecture as a data flow, steps 1–9 from DOM to speaker. Cylinders are the persistent stores: the on-device IndexedDB vector store, searched and upserted at step 2, and the Postgres cache index inside the generation service, looked up at step 5 and upserted after step 7, with the clips themselves in a separate object store. The zero-shot stage can *abstain*, making its two thresholds a dial between accuracy and exposure. Dashed red edges are the only ones carrying page text off the device; a local ONNX backend for the zero-shot stage (18-page subset, not shown) does not.

hold for five seconds before committing – the largest term in end-to-end latency (§V-F), by design.

C. Generation, loop synthesis and caching

The generation service validates the request and looks up a Postgres cache index (steps 5–7 in Fig. 1; Supabase-hosted in production, a local container for every measurement here) keyed on mood/style/key exactly and energy/valence/arousal/tempo/duration by bucket, excluding the

seed. A hit returns the stored URL and loop point; a miss generates, post-processes, writes the clip to object storage and upserts the index row. Any stage failure diverts to a pre-generated per-mood clip. Requests queue on a single batching worker; a pre-warm grid of 99 combinations is written into the index ahead of time.

Loop synthesis. The generator produces a one-shot clip, not a loop. Chroma features feed a self-similarity curve [11] over candidate loop lengths ≥ 3 s; the best candidate snaps to a bar boundary from beat times [12] (4/4 assumed), is truncated there, and its head and tail blended with a 50 ms equal-power crossfade. Loudness is normalised to -18 LUFS [13] and exported as Ogg/Opus [14] rather than MP3 – the container, not the crossfade, makes the loop gapless, since an MP3 encoder pads every loop audibly. The service also computes a diagnostic seam discontinuity, the objective measure in §V-I.

D. Extension shell and playback

Playback lives in an offscreen document: two alternating decks over raw audio buffers honour the loop point, feeding a gain and effects chain. *Fallback-then-swap* decouples audible start from generation latency: the client requests the fallback clip and a generation in parallel; the fallback plays immediately and the generated clip crossfades in when ready, with an abort controller so a new mood cancels a stale generation. *Restraint.* Sensitive pages produce silence, short-circuited before either hosted stage *when the detector flags them*; an audible active tab mutes, a background one ducks; payment, banking and browser-internal URLs bypass entirely. *Telemetry* is a local-only ring buffer, hashed and never transmitted unless explicitly exported.

V. RESULTS AND DISCUSSION

A. Evaluation design and provenance

The evaluation is organised as five studies: a systems benchmark, a cascade ablation, an audio and loop quality study, a within-subjects perceptual study, and an in-the-wild deployment study. The systems benchmark, the cascade ablation, the objective half of the audio and loop quality study, the retrieval baseline and the restraint audit are reported here; the perceptual study and the in-the-wild study await ethics approval, since both require exposing human participants to generated audio rather than replaying logged or synthetic traffic. *Stubs.* Extraction (Table III) stubs embedding inference; client latency (Table V) stubs the generation service and classify proxy. Every number is emitted by a named script against a metric registry that rejects undeclared metrics.

Corpus. The cascade-ablation corpus targets 260 pages in seven slices (182 general, 15 vocabulary-mismatch, 15 non-English, 15 mixed-genre, 12 sensitive, 15 near-miss, 6 bypass). Three derived counts do not share one arithmetic: the extraction snapshot captured 255 of 260; the ablation snapshot scores 254 with a category label; and the annotator-majority re-scoring (§V-H) covers the 208 pages where both overlap and the vote resolves. *Adversarial slice.* The restraint audit

TABLE II
AGREEMENT, 255 UNITS.

Label	α	95% CI	Ceiling	$\geq 0.667?$
Category	0.635	[0.578, 0.689]	0.621	×
Mood	0.195	[0.149, 0.238]	0.174	×
Sensitive	0.419	[0.301, 0.530]	0.775	×

TABLE III
EXTRACTION COST, 105 SITES (EMBEDDING STUBBED).

Stage	p50 (ms)	p95 (ms)	max (ms)
Text extraction	11.0	55.5	—
Colour histogram	3.9	9.7	—
Readability	1.5	7.1	—
Behaviour snapshot	0.13	0.30	—
Embedding (stubbed)	1.2	2.6	—
Vector-store search	50.3	1,222	—
Total	85.9	1,303.6	5,139

uses a separate 47-page slice (29 sensitive, 18 not), hand-written from the detector’s own term lists – white-box test content, not a naturalistic sample. *Annotators.* Three of the authors, labelling independently under a preference framing, no calibration round. *Hardware.* All measurements ran on one CPU machine except the prompt ablation (§V-J), run on a separate GPU-hosted instance (Apple M4) reached over a tunnel.

B. Annotation agreement and human ceiling

Three annotators labelled 255 pages for category (13-way), mood (11-way) and sensitivity (binary); Table II reports Krippendorff’s α and a leave-one-annotator-out ceiling. With three annotators a disagreement between the other two is a tie, counted as a miss, so the ceiling sits below α for mood and above it for binary sensitivity (ties are impossible, and the 87.6 percent skew inflates raw but not chance-corrected agreement).

All three fall below the 0.667 minimum. At $\alpha = 0.195$ for mood, annotators are near chance; Russell’s circumplex [7] gives $\alpha = 0.121$ or 0.300 under NRC norms [16] – coarsening does not reliably help, so mood is a stylistic preference, not ground truth. Mood classification accuracy is not evaluated in this paper; Table VI reports category only, though mood drives the music.

C. Extraction cost

Table III reports descriptor construction over 105 real web-sites in a Playwright-driven Chromium.

The descriptor is cheap at the median, 86 ms against a 1,989 ms navigation time; p95 is fifteen times the median, driven by the vector-store search. Boilerplate removal retained 80.8 percent of visible words. With the embedding stubbed, 86 ms is a floor. Table V reports 464 ms for a related extraction figure, but that harness measures cumulative telemetry over multiple extraction cycles on synthetic fixtures rather than a single live-site pass; the two numbers are not comparable.

TABLE IV
GENERATION LATENCY BY REQUEST CLASS, $n = 144$, NEAREST-RANK PERCENTILES. THE 12 TIMEOUTS ARE RIGHT-CENSORED AT THE CLIENT’S 300 s LIMIT.

Request class	n	p50 (ms)	p95 (ms)	max (ms)
Fallback endpoint	33	7	153	154
Cache hit	93	2,090	2,114	2,169
Generation completed (10–20 s clips)	3	139,318	207,468	207,468
Generation failed → fallback (5 s clips)	3	126,733	239,516	239,516
Client timeout, 300 s (28 s clips)	12	≥ 300,000	—	—

D. Which signals earn their cost

A per-signal offline ablation over 255 pages replays the keyword stage with one signal removed at a time. Category never changes (text-only by construction). For mood, removing colour changes the label on 7.5 percent of pages, text-alone on 10.2 percent. Behaviour and the embedding are zero *by construction*, not by finding: the corpus carries only a fixed synthetic behaviour profile, and the embedding is not an input to keyword-stage mood at all, so their live contribution is untested.

E. Generation latency and its failure modes

Table IV reports 144 requests in five distinct modes, so a mean would describe nothing that ever happened.

The fallback clip is served in a median of 7 ms. Generation completed on only three requests (82, 135, 203 s for 10, 15, 20 s clips); three more failed over to the fallback after a median of 127 s. *The twelve timeouts.* Every one is the *first* request for a 28 s clip of an uncached mood; the service’s single batching worker queued repeat attempts behind the still-running first, so later requests for the same mood only hit cache because the server finished after the client gave up. These 28 s generations are right-censored at 300 s, not early failures: at the measured real-time factor (Fig. 3) a 28 s clip sits at 230–280 s before post-processing, so 139 s from shorter clips is a lower bound on the 28 s cold cost. *Cache path.* A 2.08 s cache check on every hit, and a comparable cost saving results, point at storage I/O rather than cache logic; cache-hit throughput scaled near-linearly to 8 concurrent requests (7.94×).

Fallback-then-swap is what makes the system usable on CPU at all; its cost is that the first seconds, and on a timed-out request every second, are generic.

F. Client-side (in-browser) latency

An end-to-end Playwright harness drives the built extension against eleven synthetic fixture pages with the generation service and classify proxy stubbed; 54 of 55 page loads reached audible playback ($n = 54$).

Nav-to-audible tracks the confidence window plus the fallback path within a second, which satisfies the latency budget of §III; the dominant term the user experiences is the window we chose, not the model. One site is a reproducible outlier at 11.2–11.3 s because its fixture forces five extraction cycles

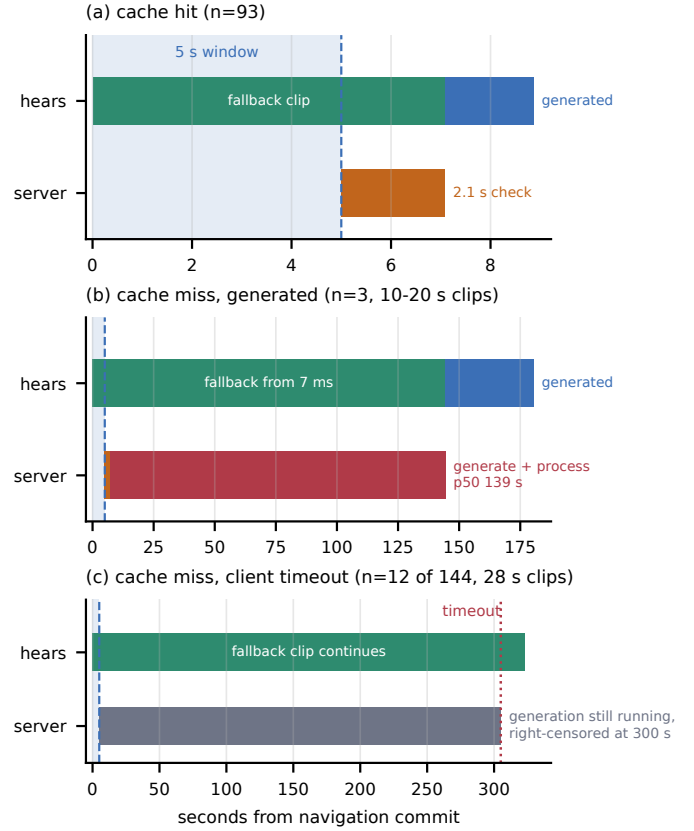


Fig. 2. Timeline from navigation to the server’s generation path, for three outcomes. (a) Cache hit, $n = 93$. (b) Completed generation, $n = 3$. (c) Client timeout at 300 s, $n = 12$, all 28 s clips: the fallback keeps playing and no swap happens.

TABLE V
END-TO-END CLIENT LATENCY (GENERATION AND CLASSIFY STUBBED), $n = 54$, NEAREST-RANK PERCENTILES.

Stage	p50 (ms)	p95 (ms)	max (ms)
Nav → page data	856	1,217	1,272
extraction telemetry (cumulative)	464	833	865
Classifier decision, per call	16	28	32
Confidence window (design constant)	5,000	5,000	5,000
Fetch + decode + crossfade start	38	74	86
Nav → audible	6,859	11,279	11,291

before a mood commits. This measures the shell only; it does not reflect the real-corpus escalation rate.

G. Cost and cache behaviour

The real corpus escalates past the keyword stage on .762 of pages – the same quantity computed slightly differently across three corpus snapshots (.756–.762), quoted here as .762 throughout – against .222 on an earlier 18-page smoke set. At that rate the cascade costs 0.0132 USD per thousand pages, negligible either way; the cascade is worth building to control disclosure, not to save money. Cache hit rate is *modelled*, not measured: 57 percent under a uniform mood distribution, 83 percent under a skewed one.

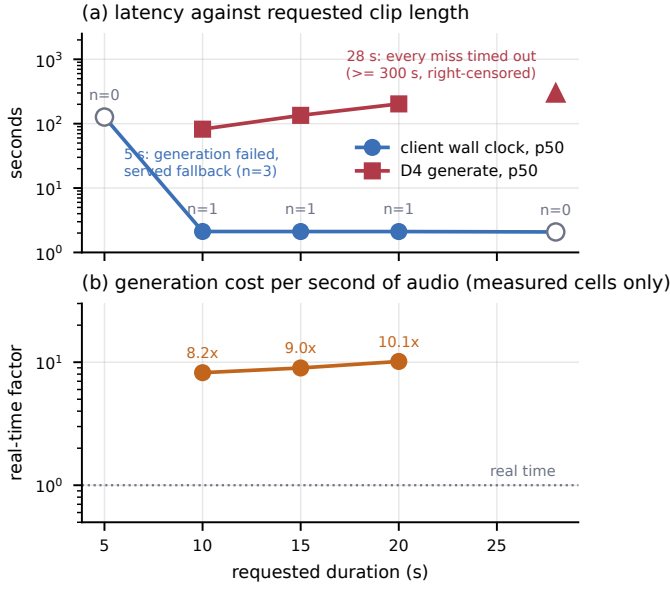


Fig. 3. Generation latency against requested duration (a) and real-time factor (b). Filled squares are the three completed generations; open markers are cells no request reached (28 s: every miss timed out, a lower bound at 300 s).

TABLE VI

CASCADE CONFIGURATIONS. ACCURACY/F1 OVER 254 LABELLED PAGES; EXPOSURE/DISCLOSURE OVER 260 RECORDS. LAST TWO COLUMNS RE-SCORE AGAINST THE ANNOTATOR MAJORITY ON 208 PAGES; A2 HAS NO RE-SCORED FIGURE.

Configuration	Acc.	F1	LLM exp.	ZS proxy	Total off-dev.	Re-scored Acc.	Re-scored F1
A1 keyword only	.189	.315	0	0	0	.197	.297
A2 zero-shot only	.520	.562	0	.938	.938	—	—
A3 LLM only	.685	.678	1.00	0	1.00	.764	.731
A4 keyword→LLM	.669	.677	.762	0	.762	.760	.705
A5 keyword→ZS→LLM	.622	.656	.250	.512	.762	.683	.683

H. Classification and the cascade

Table VI reports the cascade ablation and threshold sweep over the ablation corpus.

The full cascade (A5) trails keyword-to-LLM (A4) by 4.7 accuracy points (7.7 under the stronger label below). A5’s LLM-only exposure (.250) looks like a reduction from A4’s .762, but its zero-shot stage is itself a hosted, off-device call: every page reaching A4’s LLM stage now reaches the zero-shot proxy in A5 first. Total off-device rate is .762 for both – the cascade decides only which provider a page’s text goes to, and A5 is dominated: lower accuracy at identical disclosure, worthwhile only once its middle stage is local, which was not run at full scale.

The threshold sweep confirms this is not particular to the shipped operating point: macro-F1 ranges only .653–.668 and LLM exposure ranges .131–.596, but total off-device rate stays within .001 of .762 throughout. *Re-scoring against the annotator majority*. Category agreement is $\alpha = 0.635$, itself below the 0.667 gate, so this is the better of two imperfect labels, not validated ground truth. Re-scored on the 208 pages where the vote resolves, the A4/A5 accuracy gap widens to 7.7 points (95% CI [2.9, 12.5], $p < 0.01$), and the disclosure

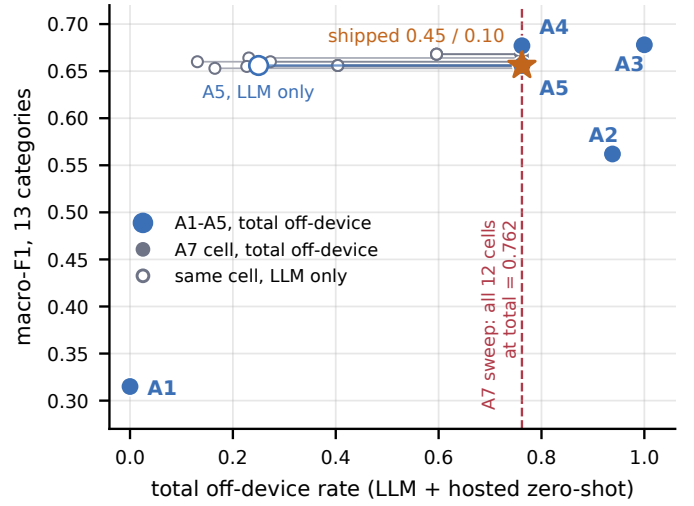


Fig. 4. Macro-F1 against total off-device rate for A1–A5 and a twelve-cell threshold sweep. Filled points (total off-device) stack at .762 regardless of threshold; hollow points (LLM-only exposure) spread .13–.60.

TABLE VII

LOOP QUALITY, 141 CLIPS, 50 MS CROSSFADE.

Measure	p50	p95
Seam energy diff., pre-crossfade (dB)	3.73	14.15
Seam energy diff., post-crossfade (dB)	2.64	10.77
Paired improvement (dB) [†]	0.93	—
Clips worsened by the crossfade		41.1%
Spectral centroid diff., pre / post (Hz)	441	557
Loudness (LUFS; target −18.0)	−18.09	—
Duration retention	0.503	0.923

[†]Median of per-clip differences, not the difference of the two medians above.

story is unchanged.

I. Loop quality

Table VII reports objective loop metrics over 141 generated clips.

Loudness and duration compliance are exact; the seam-continuity claim is not supported. *The crossfade helps less than expected*. Median paired improvement is 0.93 dB, and the crossfade worsens the energy seam on 41.1 percent of clips; on spectral centroid it is worse than nothing at the median (441 Hz before, 557 after). On a secondary sample of ten fallback clips, widening the crossfade to 500 ms only reduced median discontinuity from 3.46 to 2.90 dB, while choosing the loop point that minimises *measured* seam rather than trusting chroma similarity cut it to 0.33 dB – suggestive that the cut point, not the crossfade, is the weaker link, though at $n = 10$, unvalidated perceptually, and not on the same clips as Table VII.

Half the requested audio is discarded. Delivered clips retain a median of 50.3 percent of requested duration: 28 s yields about 14 s. The smallest retention (10.6 percent) is not a bug – the source was shorter than the 3.0 s minimum loop, so the whole clip looped. Audibility is unanswered: a listening harness (22 seeded pairs) has been validated only against simulated listeners. The self-similarity cut, then, discards half

TABLE VIII
PROMPT ABLATION, $n = 15/\text{CONDITION}$, 15 S CLIPS. OBJECTIVE METRICS ONLY.

Prompt	chars	$ \Delta E $ dB	$ \Delta \text{cent} $ Hz	Reten.
P1 bare mood word	5	3.75	565	0.527
P2 service template	175	2.63	509	0.444
P3 engineered (shipped)	312	4.35	316	0.393

TABLE IX
SENSITIVE-CONTENT AUDIT, 47-PAGE SLICE.

Metric	Result
Slice: sensitive / non-sensitive	29 / 18
False-negative rate	19/29 = .655
False-positive rate	3/18 = .167
Silence delivered on sensitive pages	34.5%

the requested audio for a seam quality it does not reliably predict; a generator conditioned to produce loop-closed material directly would remove the problem rather than patch it.

J. Does the engineered prompt earn its cost?

The shipped prompt is a 300-character engineered description. Table VIII holds mood, duration and seed fixed and varies only the prompt: P1, the bare mood word; P2, the service’s own template; P3, the engineered prompt production sends. Five moods, three generations each, no errors, no fallbacks.

On these metrics the engineered prompt does not earn its length: worst energy seam, lowest retention, best only on spectral-centroid seam. Prompt length did not change generation time. This is objective-only at $n = 15/\text{cell}$ – a prompt can lose on every number here and still be preferred by a listener – and is reported as a negative result pending the listening study, not as grounds to change the prompt.

K. Restraint policies

Fourteen executable behavioural policies – sensitive-page silence, mute/duck around existing audio, payment and browser-internal bypass, a hashed-URL telemetry guarantee, and seven direct-assertion unit cases – all pass their mechanism-level tests; the predicate behind the first of them does not. Table IX audits the sensitive-content detector on a 47-page adversarial slice.

The failures are systematic: 7 of 19 misses use vocabulary outside the term list (addiction, miscarriage, terminal illness, overdose, and similar); 4 are euphemism; 4 are non-English, which the English-vocabulary term lists cannot match (the check runs before any language gate, so this is a coverage gap, not an ordering one); 4 mention only one of two required ambiguous terms. All three false positives are near-miss academic content (a grief-counselling degree programme, a psychology textbook, an engineering article using “trauma”), unchanged as the slice grew. *Consequence.* A missed sensitive page is not silenced – it goes through the cascade like any other, reaching a hosted stage at the corpus-wide .762 rate. The disclosure constraint of §III – zero for sensitive pages – holds only for the 34.5 percent the detector flags. This is a negative result

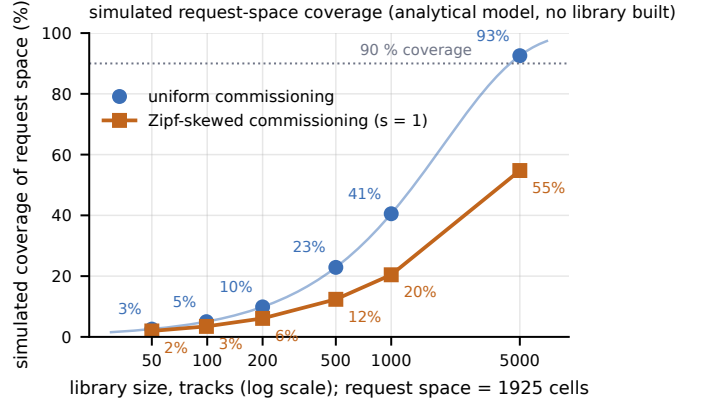


Fig. 5. Simulated coverage against library size, uniform and skewed commissioning. Analytical, not empirical.

invisible to any evaluation that tests only the mechanism; the remedy is a classifier, not more keywords.

L. Generation against retrieval

The request space is $11 \times 7 \times 5 \times 5 = 1,925$ cells. Fig. 5 is an analytical model, not a measurement: no library was built. Under a uniform random-library model a 500-track library covers under a quarter of the space and roughly 5,000 tracks are needed for 90 percent, falling to 55 percent under skewed commissioning. The claim this supports is narrow – coverage grows slowly with library size under this model – not that generation covers a space retrieval cannot: a curated 1,925-track library covers 100 percent by construction, and exact-cell matching overstates what is needed when annotators agree on mood at only $\alpha = 0.195$.

VI. CONCLUSION

Web2Music measures where its own deployment assumptions fail. A four-signal descriptor costs 86 ms median, of which only colour is shown to move the mood label offline. A three-stage abstaining classifier costs about 1.3 cents per thousand pages but redistributes disclosure rather than reducing it, and is dominated by the simpler keyword-to-LLM design at identical disclosure. Fallback-then-swap decouples a 6.9 s audible start from a generator that took a median of 139 s on the three requests that completed, with 12 of 144 requests timing out. Loop synthesis is gapless but not seamless: the crossfade worsens 41.1 percent of seams and half the requested audio is discarded to find a loop point. The sensitive-content detector misses 65.5 percent of an adversarial slice, so most sensitive pages in it would still reach a hosted model. Three annotators agree on mood at only $\alpha = 0.195$. The engineered prompt beats neither a bare mood word nor the service’s own template on most objective loop metrics. The central perceptual claim – that page-conditioned music fits the page better than page-agnostic music – remains untested.

VII. LIMITATIONS AND FUTURE WORK

The untested perceptual claim has a concrete instrument, pending ethics approval: a within-subjects forced-choice comparison of matched against shuffled audio, counterbalanced,

with a Likert fit rating and a behavioural measure (dwell time, skip/mute) alongside it. A null result would contradict this paper’s premise.

The measurements carry firm limits. Generation latency rests on three completions; no 28 s cold generation finished inside the client timeout. Client latency and the extraction embedding are both stubbed, so their figures are floors. The signal ablation is offline and keyword-stage only, uninformative by construction for behaviour and the embedding. Mood classification is unevaluated – every accuracy figure is for category while mood drives the music – and the annotators were three authors with no calibration round. The retrieval comparison assumes a random library and unverified generator prompt adherence. Loop-point selection is optimised on the metric it is evaluated on, at $n = 10$. The sensitive detector’s 65.5 percent miss rate, including all non-English cases, is a defect to fix with a model-based predicate – plausibly the entailment stage already in the extension – before any deployment, not a limitation to note in passing. The local classification backend that could reduce rather than redistribute disclosure has been measured only on 18 pages against a corpus that escalates at .762: the highest-value item remaining. The larger loop-synthesis opportunity is conditioning the generator to produce loop-closed material directly; the same descriptor-to-profile machinery would apply to other continuous contexts such as documents or code editors.

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